

CS 105: Department Introductory Course on Discrete Structures

Instructor : S. Akshay

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Lecture 01 – Introduction

Welcome to CSE@IIT Bombay!

Logistics

Course hours: Slot 10;

Tue 14:00-15:25, Thu 14:00-15:25

Office hours: To be announced.

Problem Solving/Help Session (Optional): One hour per week, run by teaching assistants. (Time and Venue to be decided)

Course material, references will be posted at

- ▶ <http://www.cse.iitb.ac.in/~akshayss/teaching.html>
- ▶ Announcements, Problem sheets and Online Discussion: [Piazza](#)
 - ▶ https://piazza.com/iit_bombay/fall2025/cs105
- ▶ One problem sheet will be posted (almost) every week!

Attendance

As per Institute rules: [SAFE](#)

More Logistics

Evaluation

- ▶ Quizzes: 30%
- ▶ Midsem: 25%
- ▶ Endsem: 40%
- ▶ Other {participation, pop quizzes, assignments}: 5%

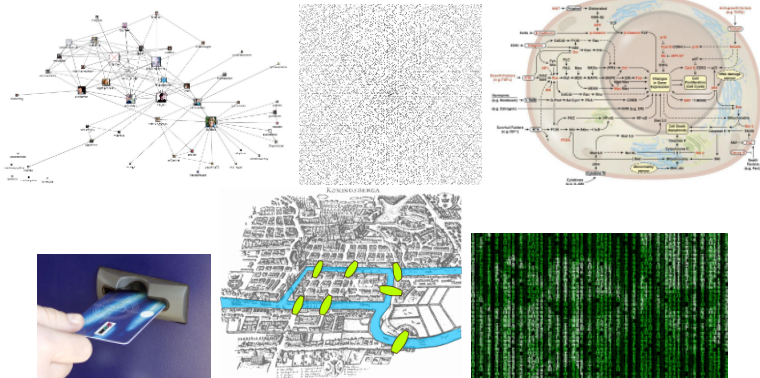
Minimum requirements (Tentative)

10/40 in endsem + 15/60 in remaining.

How to reach me after class?

- ▶ Send a message on piazza
- ▶ Drop by my office...
 - ▶ CS 507 (5th floor of New CSE/CC building)
 - ▶ Temporarily CC 313 (3rd floor!)

Goal



First things first...

- What are discrete structures?
- Why are we interested in them?

Course Outline

What we will broadly cover in this course

1. Mathematical reasoning: proofs and structures
2. Counting and combinatorics
3. Elements of graph theory
4. If time permits: Selected topics: e.g, abstract algebra and/or number theory

What we don't cover

1. Logic : predicate, first-order logic– CS228
2. Discrete probability – CS215
3. Algorithms – CS218
4. Data structures – CS213 and CS293
5. Automata theory – CS310
6. Details and applications of everything above – rest of your (academic) life!

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1. Mathematical reasoning: proofs and structures
2. Counting and combinatorics
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Textbooks

- ▶ Discrete Mathematics and its Applications with Combinatorics and Graph Theory, by Kenneth H Rosen.
- ▶ Discrete Mathematics by Norman Biggs.
- ▶ More will be listed on webpage as we go along.

More lofty aims of the course

1. Introduce **mathematical background** needed in various branches of computer science.
2. (New and old) techniques for **problem solving**: how to attack problems that you have never seen before.
3. To convey your ideas clearly: **argue logically** and **write proofs** formally.
4. To **learn abstractions** and **abstract reasoning**.

Prerequisites

- ▶ Nothing! ... Well, high school mathematics
- ▶ Logical mind and critical thinking

Chapter 1: Proofs and Logical reasoning

Outline of next few classes

- ▶ Propositions, statements
- ▶ What/why of proofs and some generic proof strategies
- ▶ Mathematical induction

Propositions

What is a proposition?

- ▶ It is raining
- ▶ $1 + 1 = 2$
- ▶ every odd number is a prime
- ▶ $2^{67} - 1$ is a prime
- ▶ $(n + 1)(n - 1) = (n^2 - 1)$ for any integer n

A **proposition** is a statement that is either true or false (but not both).

Give an example of a statement that is not a proposition.

- ▶ $x + 1 = 8$

Propositional calculus



Figure: Aristotle (384 – 322 BCE)

- ▶ propositions are statements that are either true or false.
- ▶ Just as we use variables $x, y, \dots$ for numbers, we will use variables $p, q, \dots$ for propositions.
- ▶ “if it is raining, it will be wet” : $p \rightarrow q$
- ▶ This is one way to combine propositions!

Propositional calculus and Boolean algebra

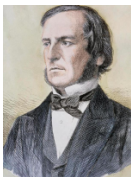


Figure: George Boole (1815 – 1864)

Combining propositions

- ▶ $\neg p$: It is not raining
- ▶ $p \vee q$: It is raining or there is a sprinkler overhead.
- ▶ $p \wedge q$: It is raining and I don't have an umbrella
- ▶ $p \rightarrow q$: If it is raining then it will be wet.
- ▶ $p \leftrightarrow q$: p if and only if q (also written p iff q)
- ▶ p, q are called atomic propositions and $\vee, \neg, \wedge, \rightarrow, \leftrightarrow$ are Boolean connectives.
- ▶ Ex: If it is raining or there is a sprinkler overhead and I don't have an umbrella, then I will get wet: $((p \vee q) \wedge r) \rightarrow s$.

Truth Tables and Logical Equivalence

p	$\neg p$
T	F
F	T

p	q	$p \vee q$	$p \wedge q$	$p \rightarrow q$	$p \leftrightarrow q$
F	F	F	F	T	T
F	T	T	F	T	F
T	F	T	F	F	F
T	T	T	T	T	T

- ▶ $\neg p$: It is not raining
- ▶ $p \vee q$: It is raining or there is a sprinkler overhead.
- ▶ $p \wedge q$: It is raining and I don't have an umbrella
- ▶ $p \rightarrow q$: If I am elected then I will lower taxes.

Logical Equivalence: Truth tables are identical!

- ▶ $p \rightarrow q$ is “same as” or logically equivalent to $\neg p \vee q$
- ▶ $p \leftrightarrow q$ is equivalent to $(p \rightarrow q) \wedge (q \rightarrow p)$
- ▶ You will pass this course iff you score $\geq 10/40$ in endsem and $\geq 15/60$ in rest.

Warning: English can be imprecise, but logic is precise!

Negation, Converse and Contrapositive

Consider the proposition: If it rains today, the match is cancelled. ($p \rightarrow q$)

1. What is its converse? ($q \rightarrow p$)
2. What is its contra-positive? ($\neg q \rightarrow \neg p$)
3. What is its negation? $\neg(p \rightarrow q)$

Which of these are equivalent to the original proposition? Why?

Exercise

Consider the proposition: If I will eat samosa or bhel puri, then I will not eat rice.

1. Write it as a Boolean combination of atomic propositions.
2. Write its converse, negation and contra-positive, both in plain english and formally.